

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS:4

Total Marks:40

1. Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.
2. Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.
3. Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.
4. Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1108

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS:4

Total Marks:40

1. Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.
2. Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.
3. Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.
4. Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

Code of Conduct:

I vishali.s(192521049) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Vishali.s

QUESTION 1:

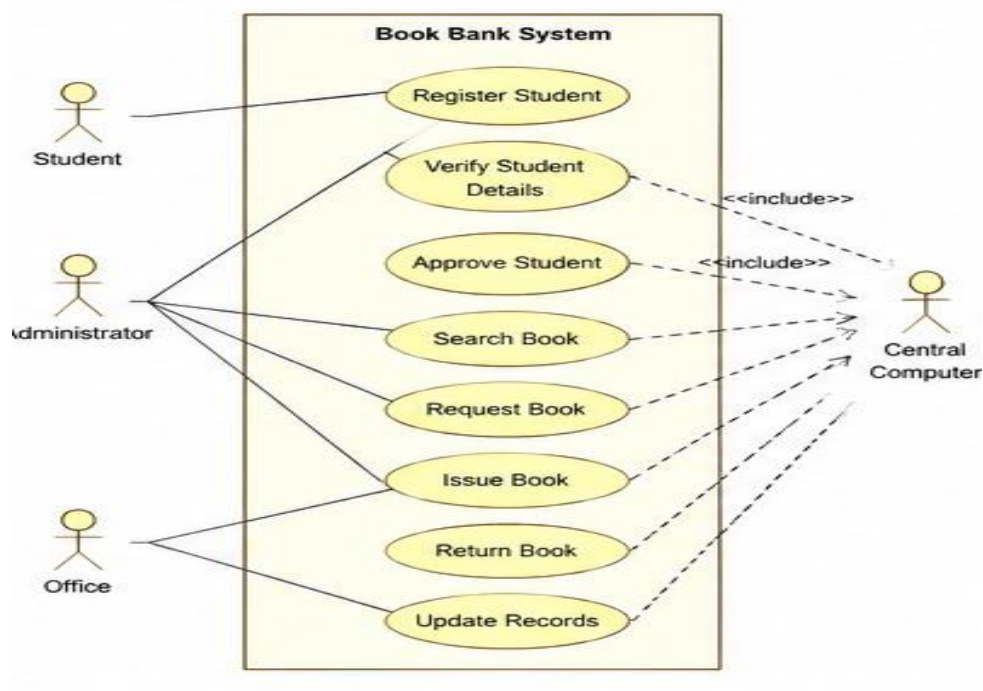
Aim

To analyze and design a Book Bank Management System using UML diagrams by identifying the system requirements, actors, classes, relationships, and interactions involved in the book issuing process.

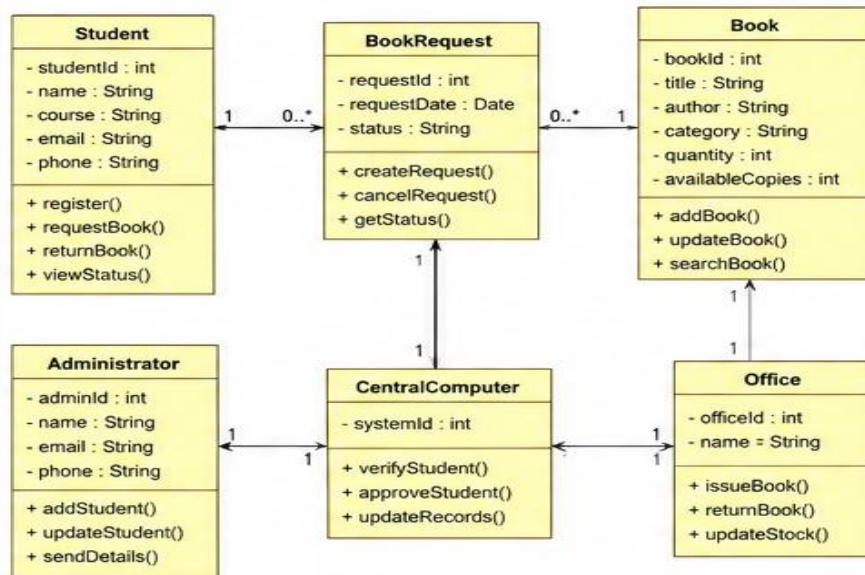
Procedure

1. Study the requirements of the Book Bank Management System.
2. Identify the actors involved, such as Student, Administrator, Central Computer, and Office.
3. Identify the use cases and the interactions between the actors and the system.
4. Design the required UML diagrams, including Use Case Diagram, Class Diagram, Sequence Diagram, Activity Diagram, State Diagram, and Object Diagram.
5. Verify that all relationships, attributes, and operations are correctly represented in the UML diagrams.
6. Analyze the system flow and ensure the diagrams accurately represent the Book Bank process.

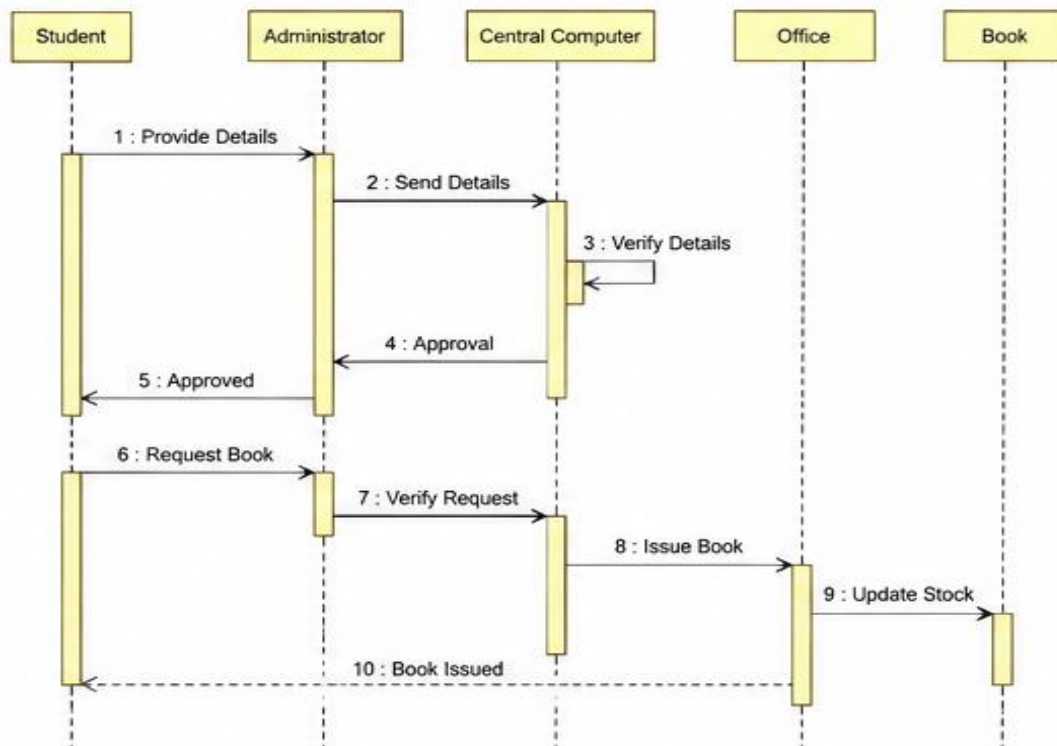
USE CASE DIAGRAM:



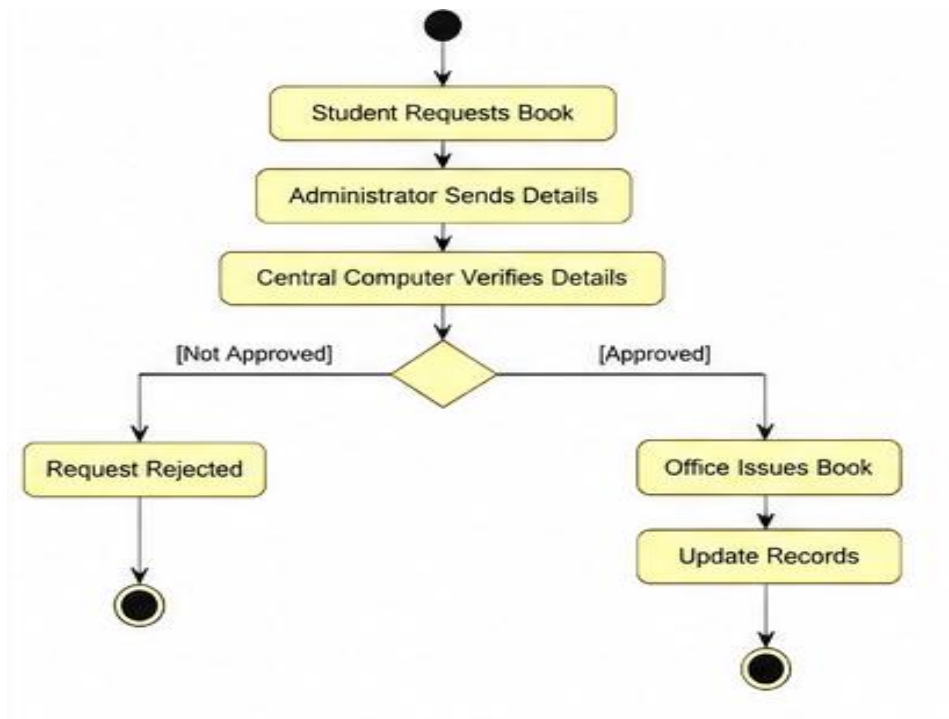
CLASS DIAGRAM :



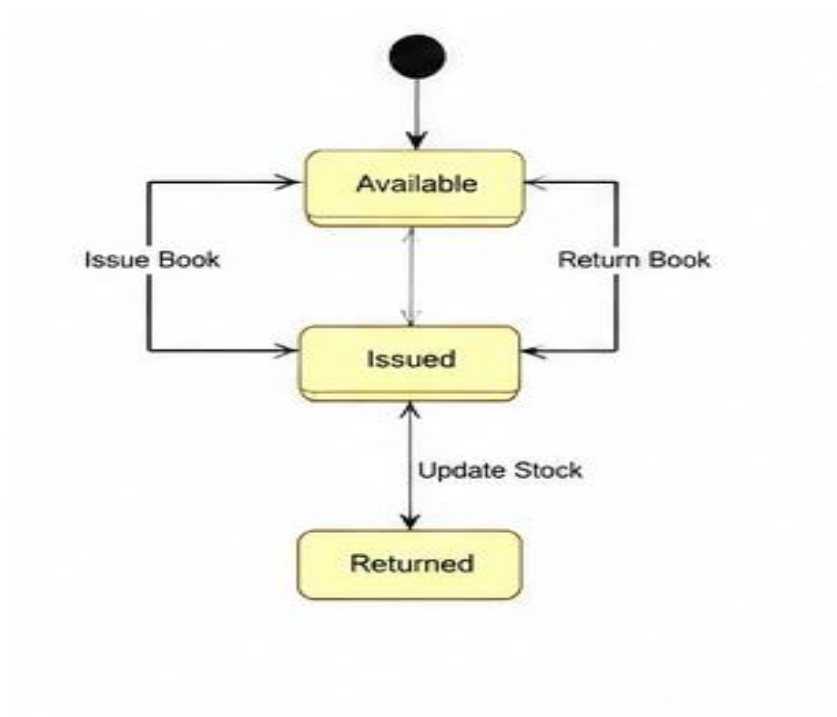
SEQUENCE DIAGRAM :



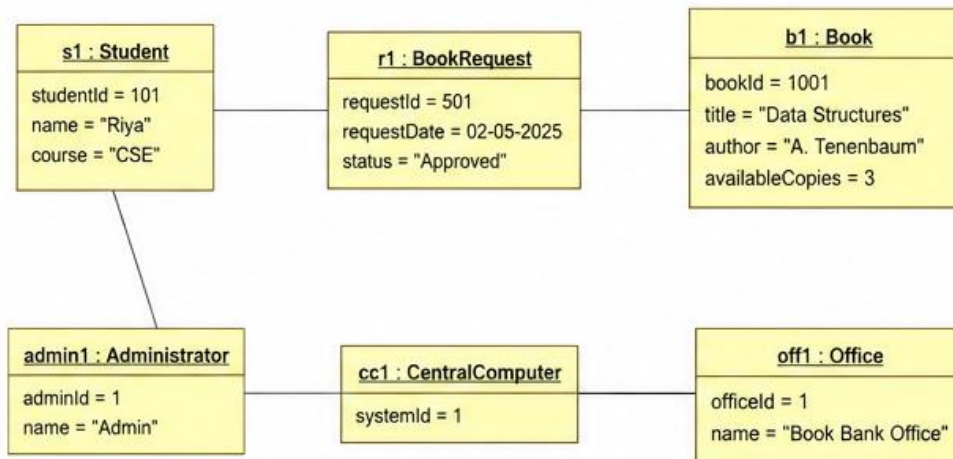
ACTIVITY DIAGRAM :



STATE DIAGRAM :



OBJECT DIAGRAM :



RESULT:

The Book Bank Management System was successfully analyzed and modeled using UML diagrams. The diagrams clearly represent the system's structure, behavior, interactions, and workflow, providing an effective object-oriented design of the Book Bank process.

This format is appropriate for your practical record and matches what is typically expected for an OOAD UML experiment.

QUESTION 2 :

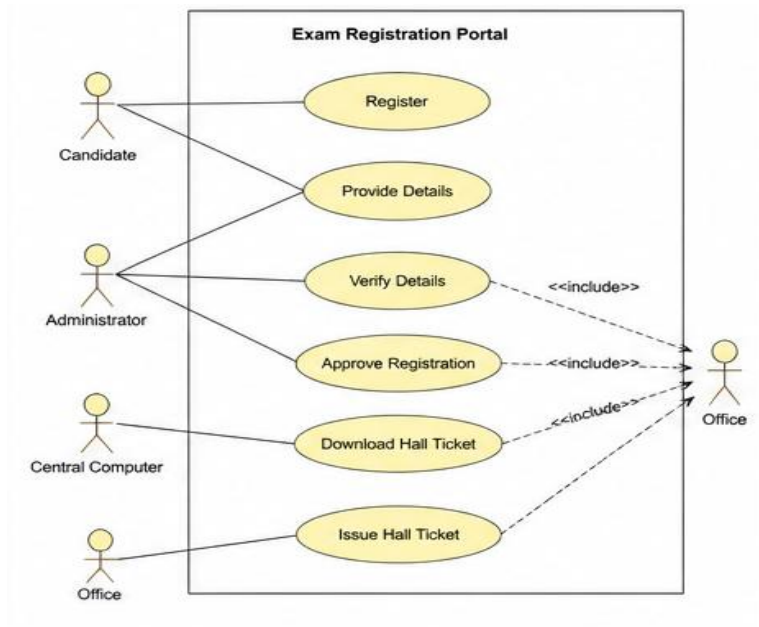
Aim:

To analyze and design an Exam Registration Portal using UML diagrams by identifying the system requirements, actors, classes, relationships, and interactions involved in the candidate registration, verification, approval, and hall ticket issuance process.

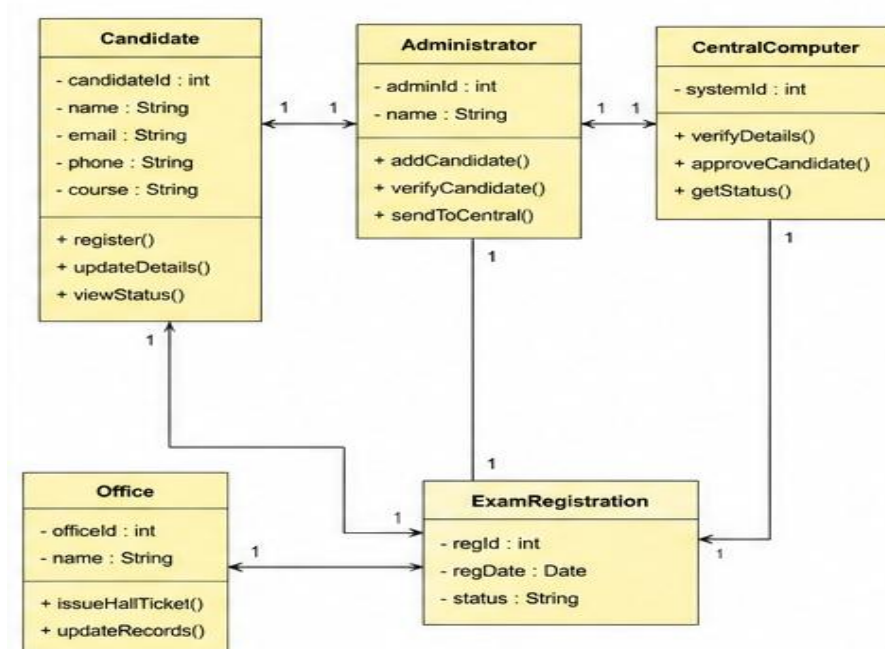
PROCEDURE:

1. Study the requirements of the Exam Registration Portal.
2. Identify the actors involved, namely Candidate, Administrator, Central Computer, and Office.
3. Analyze the functions of the system, including candidate registration, verification of candidate details, approval, and hall ticket issuance.
4. Design the required UML diagrams such as Use Case Diagram, Class Diagram, Sequence Diagram, Activity Diagram, State Diagram, and Object Diagram.
5. Verify that the candidate details are sent to the Central Computer through the Administrator and that the approval process is correctly represented.

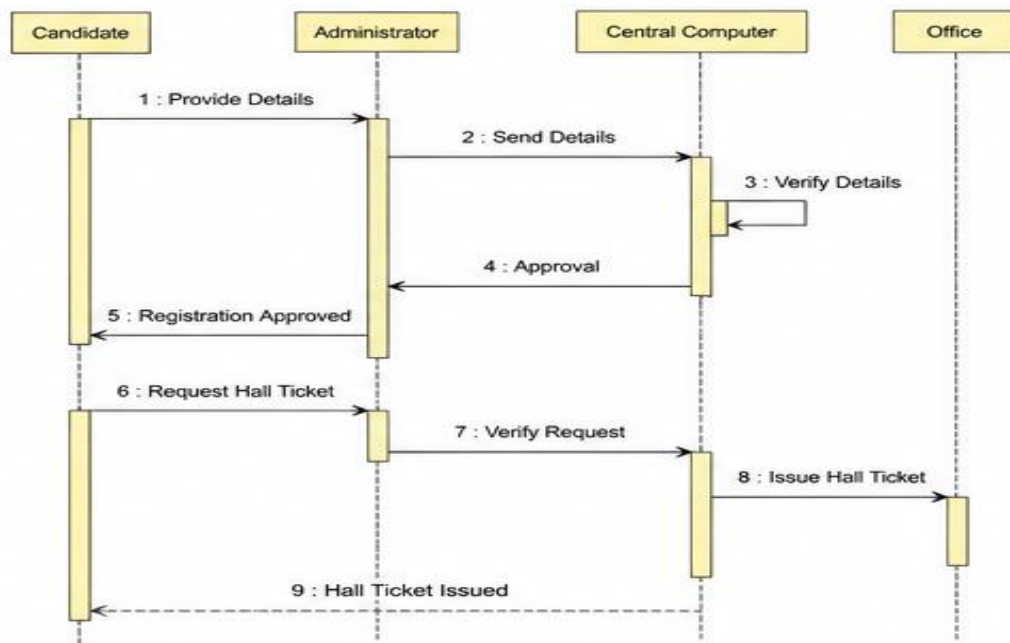
USE CASE DIAGRAM :



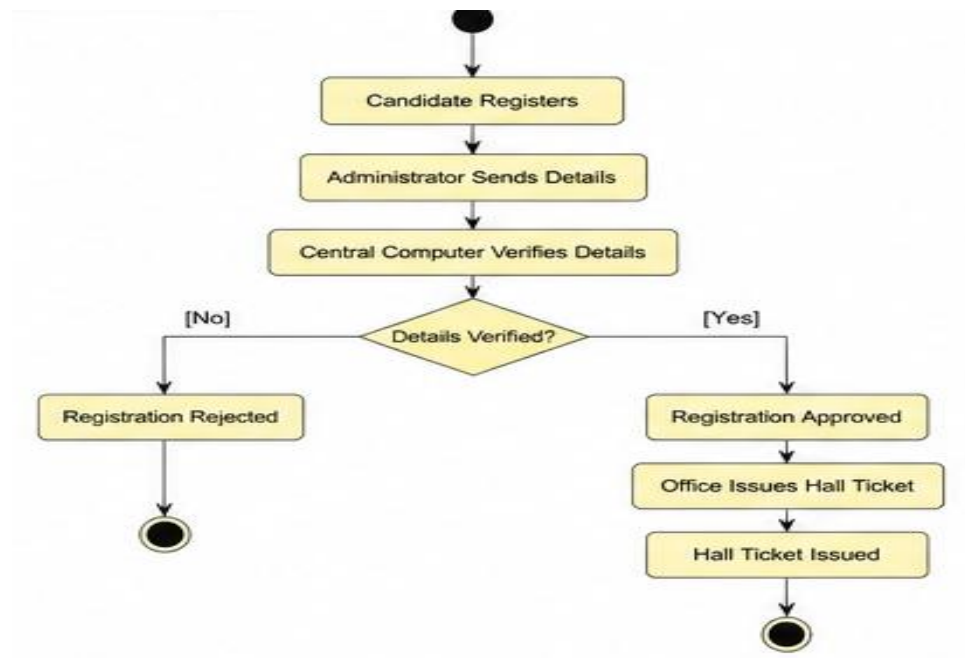
CLASS DIAGRAM :



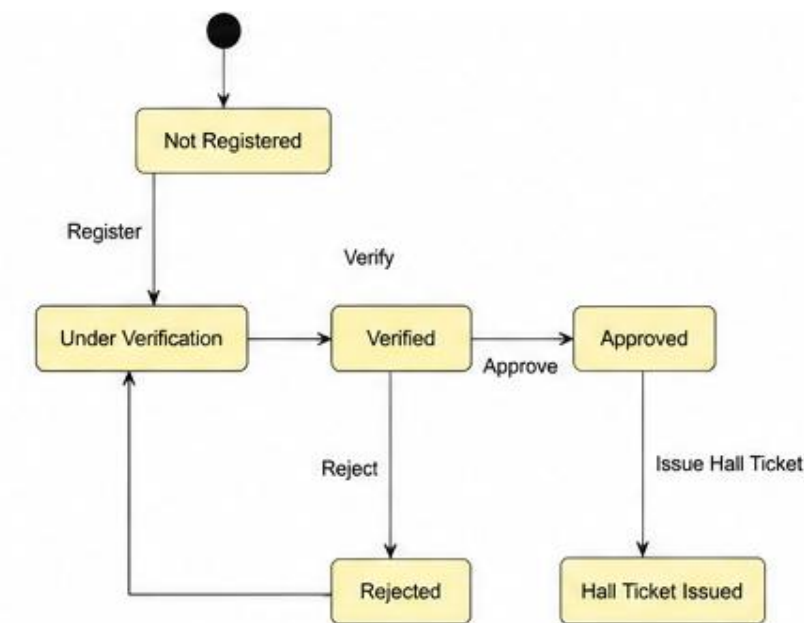
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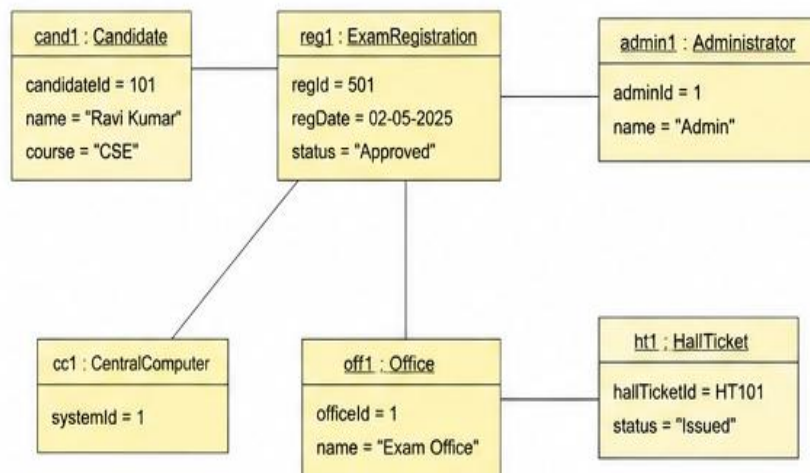
ACTIVITY DIAGRAM :



STATE DIAGRAM:



OBJECT DIAGRAM :



RESULT :

The **Exam Registration Portal** was successfully analyzed and designed using UML diagrams. The developed diagrams effectively represent the candidate registration process, verification of candidate details by the Central Computer through the Administrator, approval of the registration, and the issuance of the hall ticket by the Office, providing a clear object-oriented model of the system.

QUESTION 3:

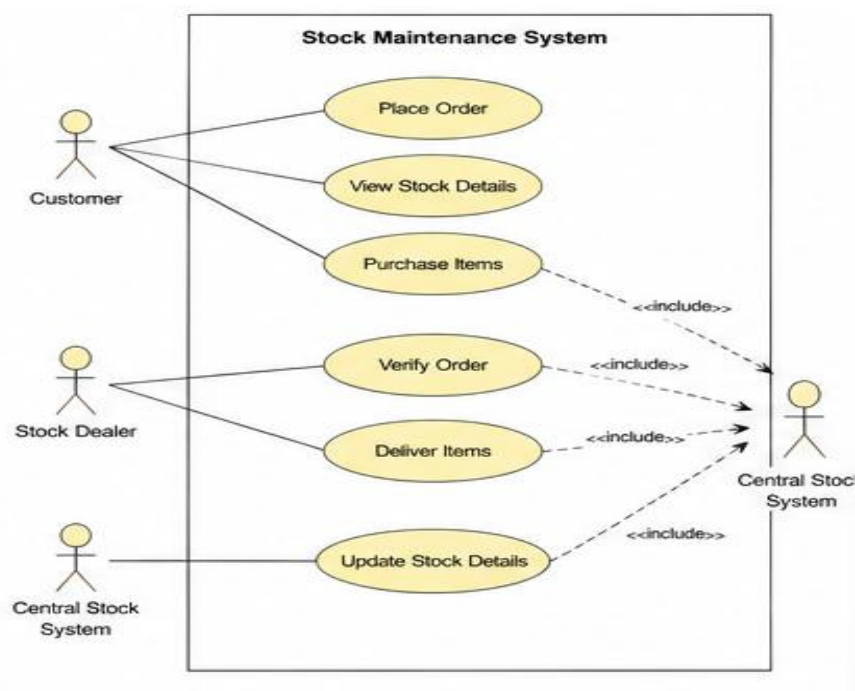
AIM

To analyze and design a Stock Maintenance System using UML diagrams by identifying the system requirements, actors, classes, relationships, and interactions involved in the order processing, stock verification, item delivery, and stock updating process.

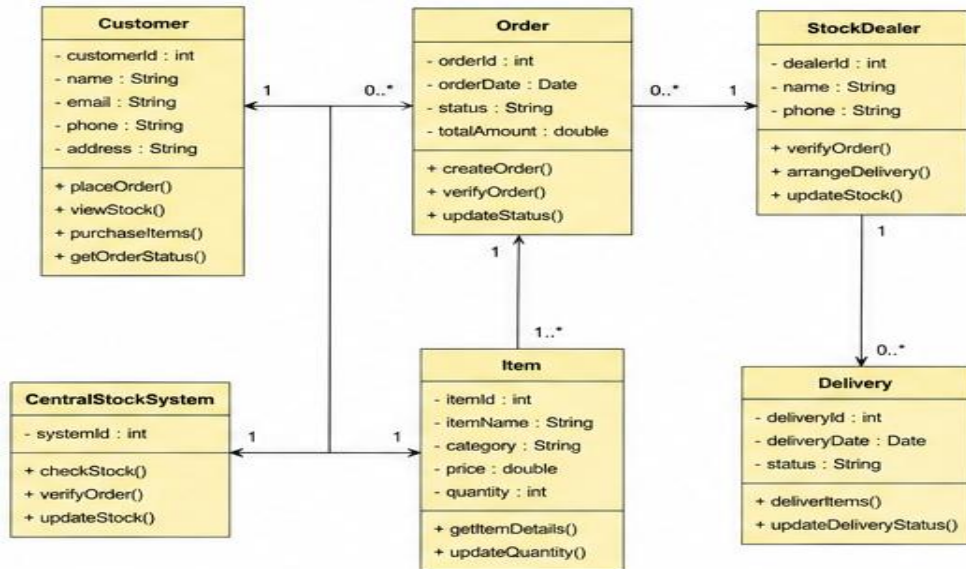
PROCEDURE

1. Study the requirements of the Stock Maintenance System.
2. Identify the actors involved, namely Customer, Stock Dealer, and Central Stock System.
3. Analyze the functions of the system, including placing orders, verifying orders, purchasing items, delivering items, and updating stock details.
4. Design the required UML diagrams such as Use Case Diagram, Class Diagram, Sequence Diagram, Activity Diagram, State Diagram, and Object Diagram.
5. Verify that the customer's order is processed through the Stock Dealer and Central Stock System before the items are delivered.
6. Validate the complete workflow from order placement to stock update using UML diagrams

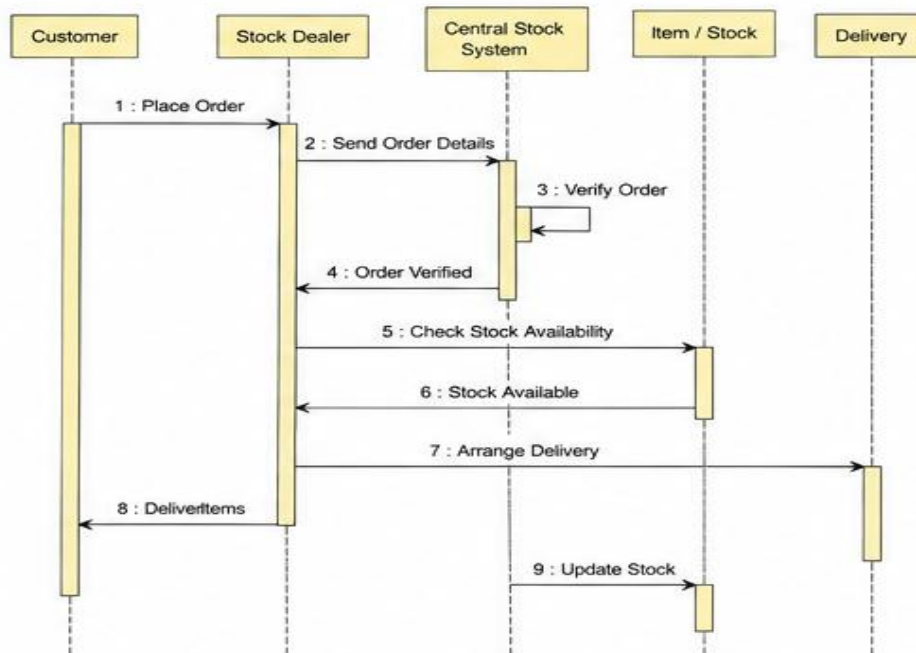
USE CASE DIAGRAM:



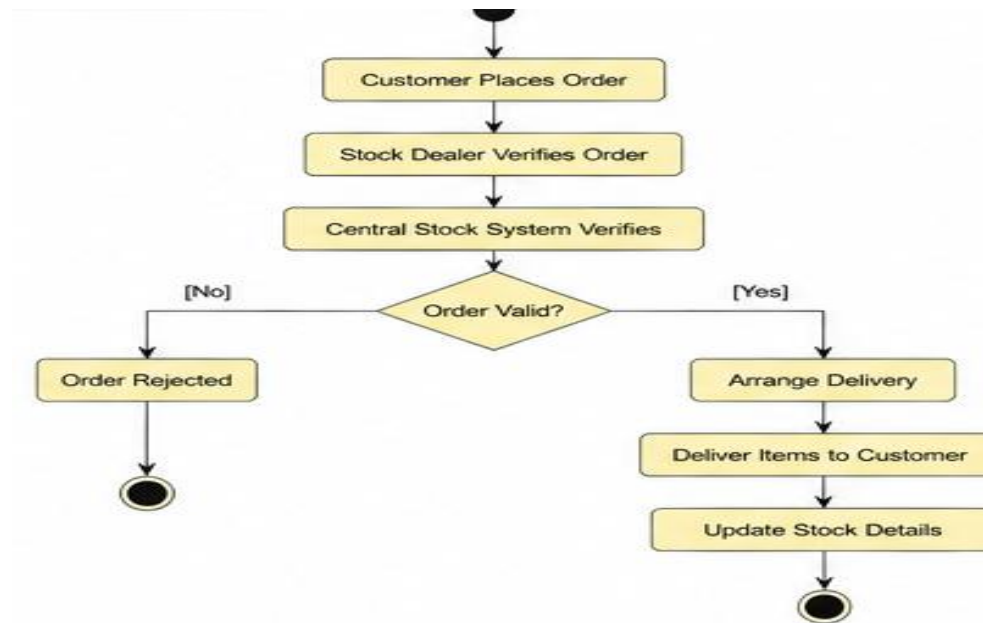
CLASS DIAGRAM :



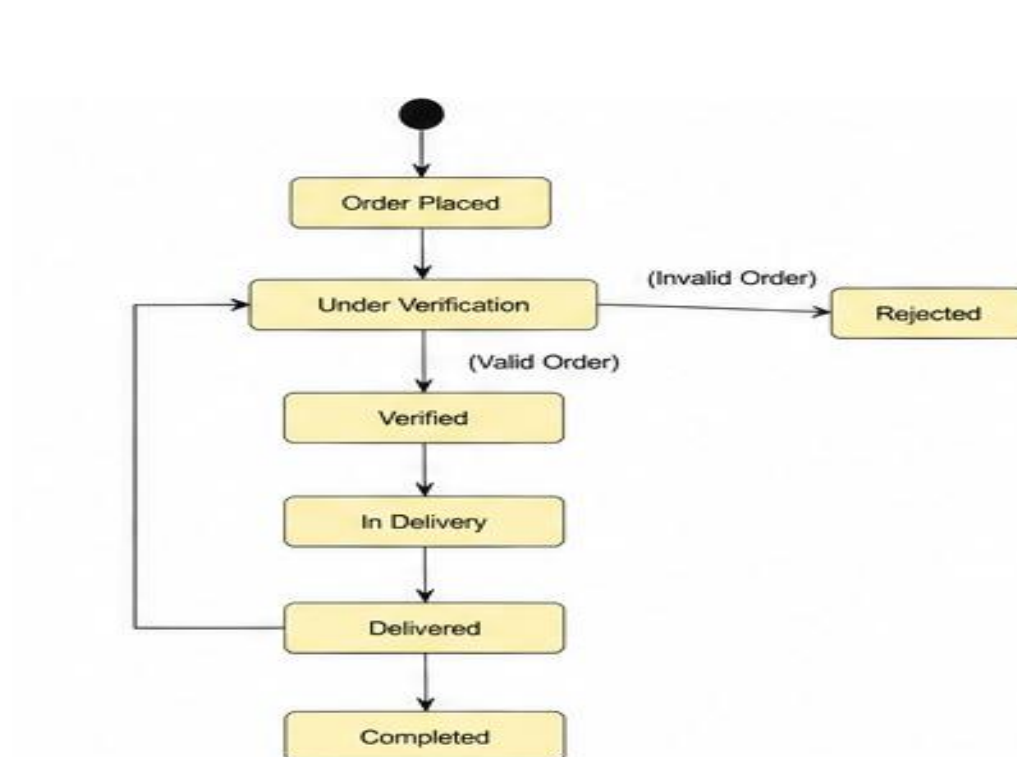
SEQUENCE DIAGRAM:



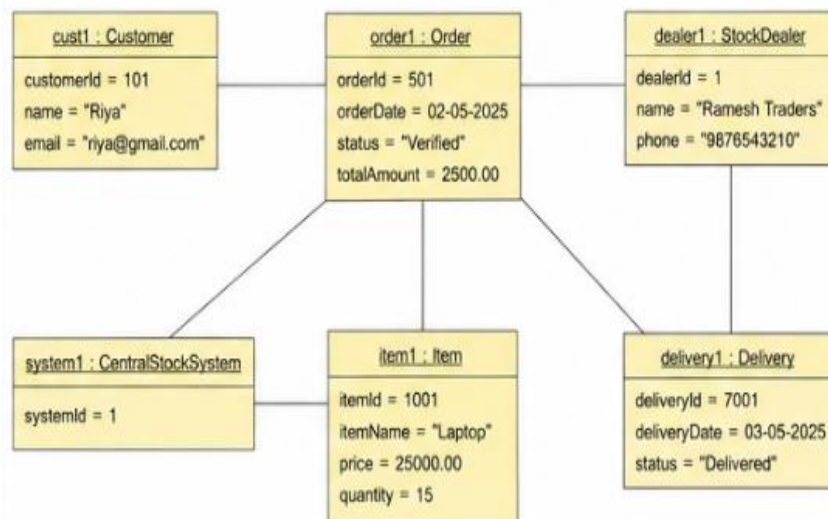
ACTIVITY DIAGRAM:



STATE DIAGRAM:



OBJECT DIAGRAM:



RESULT :

The Stock Maintenance System was successfully analyzed and modeled using UML diagrams. The diagrams clearly represent the order placement, order verification, item delivery, and stock updating processes, providing an effective object-oriented design of the Stock Maintenance System.

QUESTION 4:

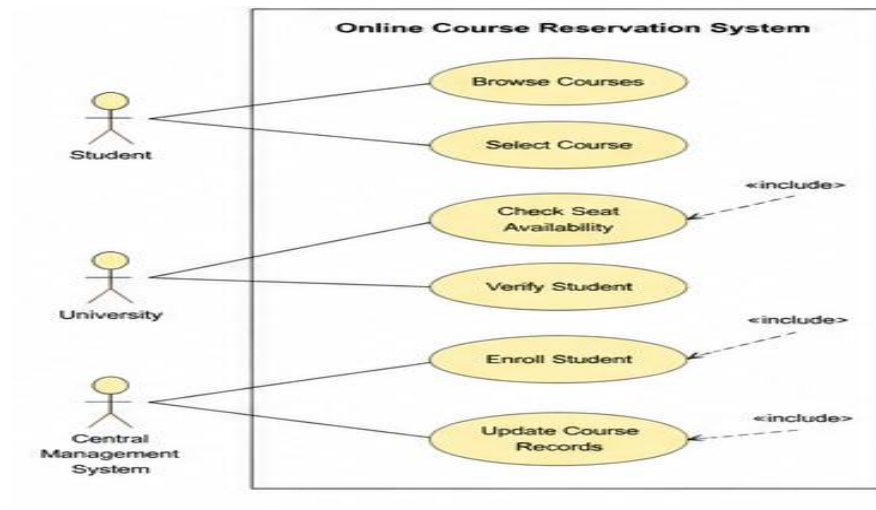
AIM :

To analyze and design an Online Course Reservation System using UML diagrams by identifying the system requirements, actors, classes, relationships, and interactions involved in course selection, seat availability checking, student enrollment, and course record management.

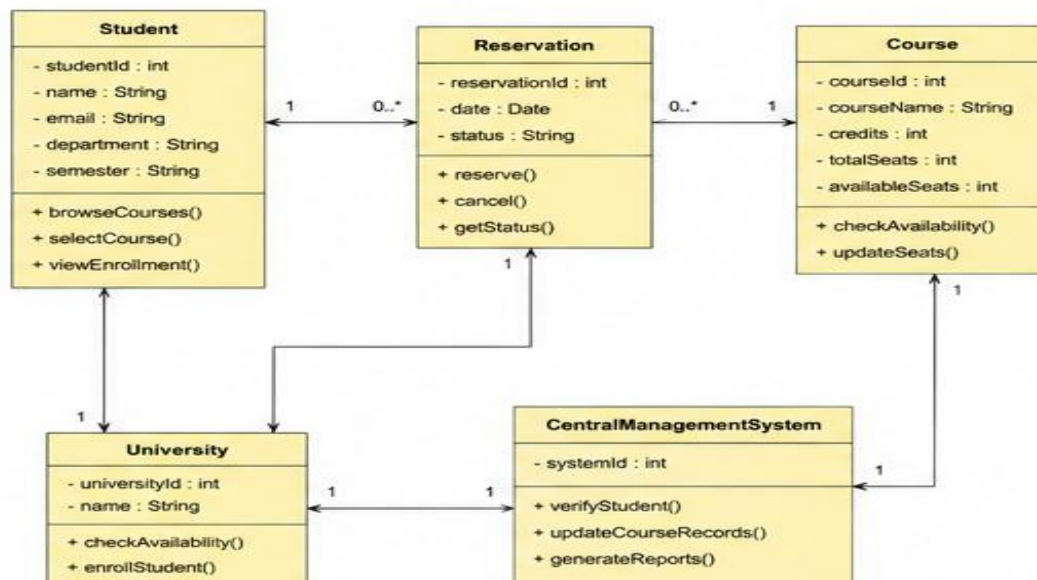
PROCEDURE :

1. Study the requirements of the Online Course Reservation System.
2. Identify the actors involved, namely Student, University, and Central Management System.
3. Analyze the functions of the system, including browsing courses, selecting a course, checking seat availability, verifying the student, enrolling the student, and updating course records.
4. Design the required UML diagrams such as Use Case Diagram, Class Diagram, Sequence Diagram, Activity Diagram, State Diagram, Object Diagram, and Component Diagram.
5. Verify that the selected course has available seats before enrolling the student through the Central Management System.

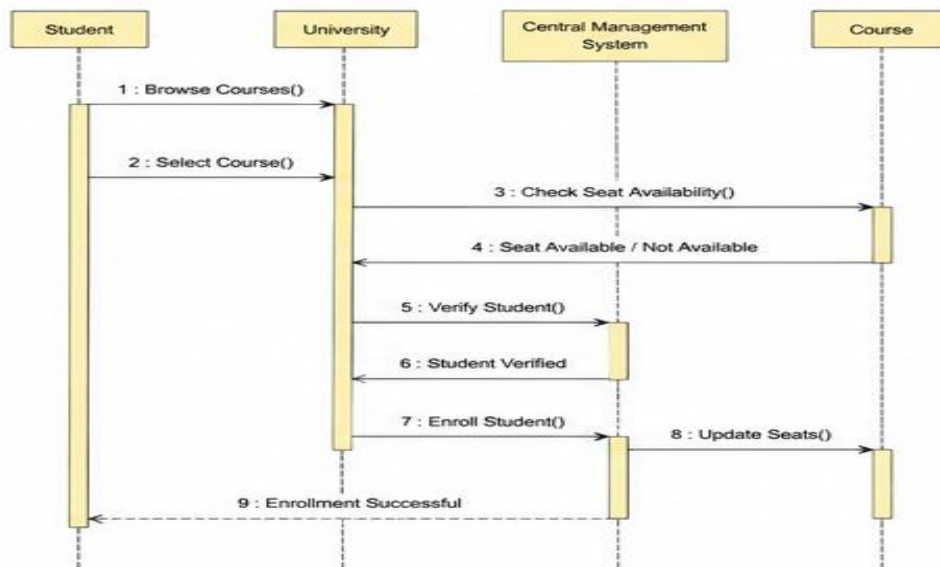
USE CASE DIAGRAM :



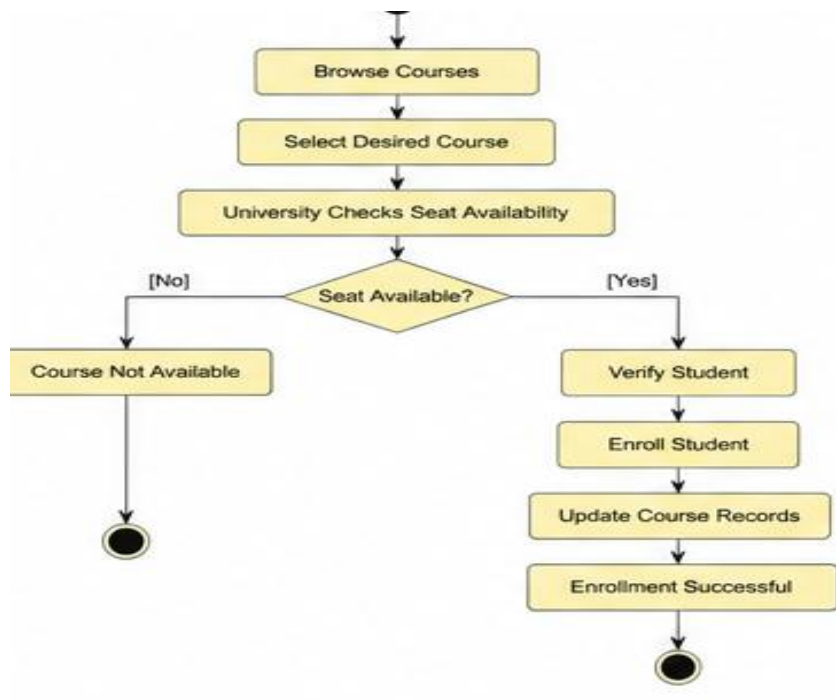
CLASS DIAGRAM :



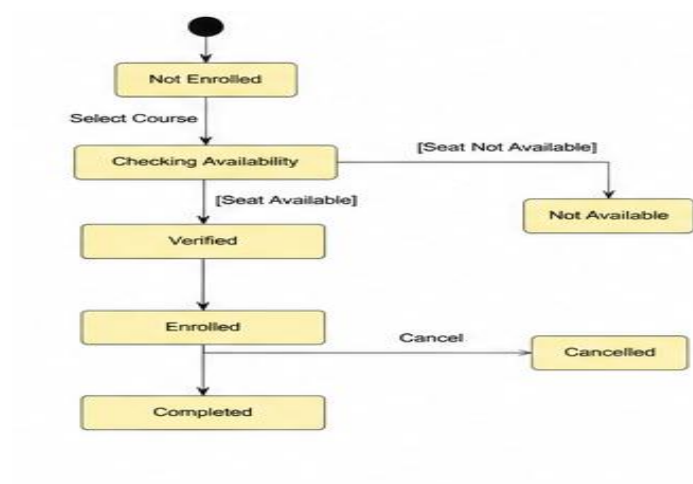
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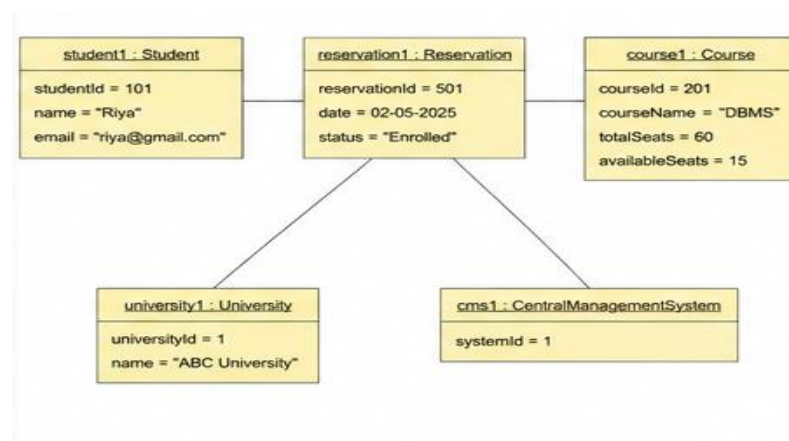
ACTIVITY DIAGRAM :



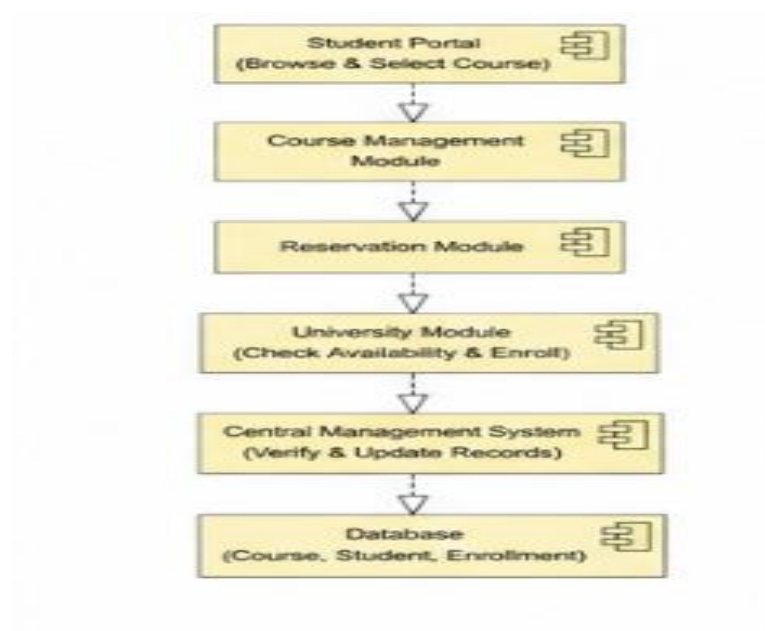
STATE DIAGRAM :



OBJECT DIAGRAM :



COMPONENT DIAGRAM :



RESULT:

The Online Course Reservation System was successfully analyzed and modeled using UML diagrams. The diagrams clearly represent the course selection process, seat availability checking, student verification, enrollment, and course record management, providing an effective object-oriented design of the system.

RUBRICS FOR CALCULATION:

Experiments

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation